Alexander Marshall | Physicist

1900 Glenacre Drive – Kernersville, North Carolina 27284
☎ (336) 618 9851 • ✉ al3xmarshall99@gmail.com
philosophiaephysica.org • Royster Doctoral Fellow at UNC Chapel Hill

HR Department
Corporation
123 Pleasant Lane
12345 City, State

February 1, 2026

To Whom it May Concern,

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Sincerely,

Alexander Marshall

Attached: Curriculum Vitæ

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Education

Doctorate of Philosophy in Physics — **Chapel Hill, NC**
The University of North Carolina at Chapel Hill — *2027 (Anticipated)*
Royster Doctoral Fellow | Exams Passed | All But Dissertation

Master of Science in Physics — **Chapel Hill, NC**
The University of North Carolina at Chapel Hill — *2025*
Completed *en route* to PhD

B.S. in Physics, B.A. in Philosophy, Minor in Math. — **Winston-Salem, NC**
Wake Forest University — *2022*
GPA 3.91 | *Summa Cum Laude* | Stamps Scholar

High School Diploma — **Clemmons, NC**
West Forsyth High School — *2018*
Junior Marshall | Top 1% of graduating class

Academic Projects

Honors Thesis

Generating a 3-D Lattice of Photonic Spheres to Track Fluorescently Labelled Chromatin

Advisor: Dr. Keith Bonin

Comittee: Dr. Keith Bonin, Dr. George Holzwarth, Dr. Stephen Baker

Description: Wrote a custom iterative routine to generate, test, and analyze hundreds of thousands of phase settings on a spatial light modulator, implementing the correction method of Dholakia et al. (2017) Used this routine to create 3D lattice of photonic spheres to be used for tracking labelled chromatin in irradiated cells.

Masters Thesis

Physiological Strains in Poroelastic Media Induce Asymmetric Transport.

Advisor: Richard Superfine

Thesis Comittee: Dr. Richard Superfine, Dr. Greg M. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Kerry Bloom

Description: Tracked quantum dots undergoing dynamic diffusion in poroelastic media using LSFM with unprecedented temporal resolution. Live imaging during physiological strain reveals that large solute transport is asymmetric during dynamic strains, with advection yeilding significantly more forward angular anisotropy during the relaxation stroke than the compression stroke. These results suggest single-particle effects could explain the macroscopic solute concentration enhancements reported by studies of poroelastic media under dynamic strain.

Doctoral Thesis

Anomalous Diffusion in Poroelastic Media

Advisor: Richard Superfine

Thesis Comittee: Dr. Richard Superfine, Dr. C.A. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Ehssan Nazockdast

Description: Explore the role of static and dynamic strains in the transport of large solutes in various poroelastic media, including transcription factor search patterns in cellular nuclei exposed to dynamic strains.

Additional Research

Chalcogenide Crystals for Quantum Computing — **Winston-Salem, NC**
Wake Forest Nanotech Center, Dr. David Carroll — *2021*

Neurochemical Addiction Mechanisms In the Nocus Accumbens — **Winston-Salem, NC**
Wake Forest School of Medicine, Dr. Evgeny Budygin — *2019-2021*

Cultural Impact of Oenology in the Peleponnese — **Winston-Salem, NC**
Wake Forest Department of Anthropology, Dr. Karen Friederic — *2019*

Cancer and Postoperative Pain — **Winston-Salem, NC**
Wake Forest Baptist Medical Center, Dr. Christopher Peters — *2018*

Scholarships and Grants

2022-2027: **Royster Doctoral Fellowship.** Five-year fellowship awarded for academic merit.

2018-2022: **Stamps Scholarship.** Four-year scholarship awarded for academic merit.

2019: **Wake Forest Scholars Travel Grant.** Funded summer anthropology project abroad.

Publications

Technical skills

Advanced Coding: MATLAB, LabVIEW, Python, LaTeX, Git

Scripting: μManager, ImageJ

Data Analysis: Particle Tracking, Image Analysis, Stochasitic Processes

Other: Transmission Microscopy, Fluorescent Microscopy, Light Sheet Fluorescent Microscopy, Total Internal Reflection Microscopy, Custom Optical Design, Laser Systems, Spatial Light Modulation, Remote Focusing, Machine Vision, Atomic Force Microscopy, general lab skills, hazardous waste management

Presentations and Symposia

Generating a Three-Dimensional Lattice of Photonic Spheres — **Winston-Salem, NC**
Symposium, Wake Forest University Physics Department — *May 2022*

Diffusion in Strained Poroelastic Media — **Chapel Hill, NC**
Thesis Proposal, UNC Chapel Hill Physics Department — *September 16, 2025*

Strain Alters Anomalous Diffusion in Poroelastic Media. — **Elon, NC**
Symposium, Elon University Physics Department — *October 1, 2025*

Physiological Strains in Poroelastic Media Induce Asymmetric Transport. — **Chapel Hill, NC**
Master's Defense, UNC Chapel Hill Physics Department — *November 17, 2025*

Conferences

Undergraduate Research Day **Winston-Salem, NC**
Symposium, Wake Forest University URECA Center *September 2019*

Strain Driven Transport In Poroelastic Media **Durham, NC**
Triangle Cytoskeleton Meeting, Duke University *September 15, 2025*

Strain Driven Transport In Poroelastic Media **Raleigh, NC**
Triangle Cytoskeleton Meeting, NC Museum of Natural Sciences *September 23, 2024*

Probing Structural and Force Dynamics during Phagocytosis **Raleigh, NC**
Triangle Cytoskeleton Meeting, NC Museum of Natural Sciences *September 25, 2023*

Various topics **Chapel Hill, NC**
Annual Colloqium, UNC Applied Physical Sciences Department *2022-2025*

Strain Driven Transport In Poroelastic Media **Raleigh, NC**
8th Inernational Soft Matter Conference, Soft Matter Association of the Americas *August 2024*

Strain Driven Transport In Poroelastic Media **Durham, NC**
Triangle Soft Matter Workshop, Duke University Physics Department *May 12, 2025*

Strain Driven Transport In Poroelastic Media **Chapel Hill, NC**
13th Carolina Biophysics Symposium, UNC Chapel Hill *November 6, 2025*

Strain Driven Transport In Poroelastic Media **Chapel Hill, NC**
Royster Global Conference, Royster Soceity of Fellows *October 21, 2025*

Strain Driven Transport In Poroelastic Media **San Francisco, CA**
Biophysical Society Meeting, Biophysical Society *February 21, 2026*

Teaching and Mentorship

2021-Present: Many semesters experiance as a TA.

2018-Present: Private tutor in advanced STEM fields.

Undergraduate Research Metnorship: 2023-Present

Outreach

Wake Forest Baptist Medical Center **Winston-Salem**
Volunteer, over 400 accumulated hours *2015–2017*

UNC APS Department **Chapel Hill**
Open Flexure Microscope and Molecular Stamper *2023-Present*
Constructing low cost microscopes for use in low-resource environments. Project has recieved multiple awards, including the UNC MakerFest Entrepreneurship Award.

UNC Chapel Hill **Chapel Hill**
UNC ScienceFest *2023-Present*

Select Coursework

Graduate Mathematical Methods *2022*

Graduate Quantum Mechanics *2022*

Graduate Classical Dynamics *2022*

Graduate Electromagnetism	*2023*
Graduate Statistical Mechanics	*2023*
Optics	*2024*
Fluid Dynamics	*2025*
Biophysics	*2025*
Partial Differential Equations	*2022*
Quantum Computing	*2022*

References

Richard Superfine, PhD: superfine@unc.edu — *Doctoral Advisor*
Kerry Bloom, PhD: kerry_bloom@unc.edu — *Reasearch Collaborator*
David Hill, PhD: dbhill@email.unc.edu — *Doctoral Committee Member*
Keith Bonin, PhD: bonin@wfu.edu — *Undergraduate Advisor*
Evgeny Budygin, PhD, PharmD: budygin@gmail.com — *Research Collaborator*

Languages

English: Fluent — *Native Speaker*
Modern Greek: Intermediate — *Conversational*
Classical Latin: Intermediate — *Several years of formal instruction*

Personal Interests

-Music
-Philosophy
-Running
-Backpacking